LifeLine — Comprehensive Blood Donation Platform

> **Document**: Vision Document
> **Course**: CSC13002 - Introduction to Software Engineering
> **Team**: Sanguine (Group 05)
> **Version**: 1.1 | **Date**: 23/06/2026

Table of Contents

Revision History

Date	Version	Description	Author
14/06/2026	1.0	FG 1.1 User Account Management, FG 1.2 Donation Booking & Location Services, Workflow Diagram: Donor Booking flow in Mermaid Features: FG 1.3 Q&A AI, FG 1.4 News & Notifications, Workflow Diagram: AI Feature flow in Mermaid Features: FG 3.1 SOS Hospital Emergency Request Management, Section 6: Non-Functional Requirements Features: FG 2.1 Campaign & Donor Management, FG 2.2 Communication & Engagement Management, FG 2.3 Blood Inventory & Emergency Coordination FG 4.1 User Automations, FG 4.2 BC Automations, Workflow Diagram: SOS Emergency flow in Mermaid Section 1, 2, 3, 4; Features: FG 1.5 Donation Impact & Tracking, FG 1.6 Community	Trần Anh Kiệt Trần Đức Quý Nguyễn Quốc Dương Trần Minh Triết Trần Minh Triết Trịnh Khánh Linh
23/06/2026	1.1	Refined Headings, Rewrote System Features, Added 5.4.3 Notification Service Added 6.5 Applicable Standards	Trần Anh Kiệt Nguyễn Quốc Dương

1. Introduction

(Author: Trịnh Khánh Linh | Reviewer: Trần Anh Kiệt | Editor: Trịnh Khánh Linh)
Features: FG 1.5 Donation Impact & Tracking, FG 1.6 Community
The purpose of this document is to collect, analyze, and define the high-level needs and features of the LifeLine platform. It focuses on the capabilities required by stakeholders and target users, as well as the reasons behind these needs. Detailed specifications of how LifeLine fulfills these requirements are described in subsequent use-case and supplementary documents.

This document provides an overview of the LifeLine project, which aims to develop a modern blood donation management and donor engagement platform. The system is designed to help blood donors, blood donation organizations, and hospitals manage donation activities more effectively while improving donor retention through personalized support, timely communication, and AI-powered assistance.

2. Positioning

(Author: Trịnh Khánh Linh | Reviewer: Trần Anh Kiệt | Editor: Trịnh Khánh Linh)

2.1 Problem Statement

The problem of	fragmented blood donation information, inefficient donor engagement, and difficulties in retaining regular blood donors
affects	blood donors, blood donation organizations, hospitals, and healthcare staff managing donation activities
the impact of which is	reduced donor participation, uncertainty regarding donation eligibility, lack of post-donation support, and challenges in maintaining a stable blood supply
a successful solution would be	a centralized platform that provides reliable donation information, personalized donor support, automated reminders, and effective communication between donors and healthcare organizations

2.2 Product Position Statement

For	blood donors, blood donation organizations, and hospitals
Who	need an efficient and engaging way to manage blood donation activities and maintain long-term donor participation
The LifeLine blood system	is a blood donation management and donor engagement platform
That	simplifies donation registration, provides personalized guidance, supports post-donation care, and improves communication between donors and healthcare organizations

Unlike	existing blood donation platforms that primarily focus on appointment scheduling and donation record management
Our product	combines AI-powered assistance, donor engagement mechanisms, community interaction, intelligent reminders, and real-time communication tools to improve donor retention and operational efficiency

3. Stakeholder and User Descriptions

(Author: Trịnh Khánh Linh | Reviewer: Trần Anh Kiệt | Editor: Trịnh Khánh Linh)

3.1 Stakeholder Summary

Name	Description	Responsibilities
Supervisor	Teaching Assistant overseeing the project	Provide guidance throughout development, review documents, designs, and source code
Development Team	Team responsible for system development	Design, implement, test, and maintain the LifeLine platform
Blood Donation Organizations	Organizations responsible for organizing blood donation campaigns	Manage donation events, donor registrations, and communication with donors
Hospitals	Medical institutions requiring blood supply management	Monitor blood inventory, issue blood requests, and coordinate donation activities
System Administrators	Personnel responsible for system operation	Manage users, monitor system performance, and maintain platform security

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Donor	Individuals who participate in blood	Register for donations, manage appointments, track donation	Blood Donation Organizations,

Name	Description	Responsibilities	Stakeholder
	donation activities	history, receive reminders and health guidance	Hospitals
Organization Staff	Staff managing donation campaigns and donor registrations	Manage events, review registrations, communicate with donors	Blood Donation Organizations
Hospital Staff	Staff responsible for blood inventory and emergency requests	Monitor blood availability, create urgent donation requests, coordinate with donors	Hospitals
Administrator	Users with full system privileges	Manage users, permissions, platform settings, and reports	System Administrators

3.3 User Environment

Donor

- Number of people: Potentially thousands of registered donors.
- Task cycle: Browse donation opportunities, register for appointments, receive reminders, and track donation history periodically.
- Environment: Home, school, workplace, or public locations using internet-connected devices.
- Platforms: Mobile browsers, desktop browsers.
- Integration needs: Email services, push notifications, and location services.

Organization Staff

- Number of people: Multiple staff members per organization.
- Task cycle: Manage campaigns daily, review registrations, and communicate with donors.
- Environment: Office-based work with stable internet connectivity.
- Platforms: Desktop and laptop computers.
- Integration needs: Notification services, donor databases, and reporting systems.

Hospital Staff

- Number of people: Medical staff and blood bank personnel.
- Task cycle: Monitor blood supply levels, manage emergency requests, and coordinate donation events.

- Environment: Hospital and blood bank facilities.
- Platforms: Desktop computers and tablets.
- Integration needs: Blood inventory systems and hospital databases.

Administrator

- Number of people: 1–3 administrators.
- Task cycle: User management, system monitoring, maintenance, and troubleshooting.
- Environment: Office environment with reliable internet access.
- Platforms: Desktop and laptop computers.
- Integration needs: Monitoring tools, backup systems, and security services.

3.4 Summary of Key Stakeholder or User Needs

Need	Priority	Concerns	Current Solution	Proposed Solution
Centralized donation information	High	Information is scattered across social media and multiple channels	Social media posts and manual announcements	Centralized information portal and event directory
Donation eligibility guidance	High	Users are uncertain about donation requirements	FAQ pages and external consultation	AI-assisted eligibility guidance and screening support
Post-donation health support	High	Donors seek recovery information from external sources	Google searches and healthcare professionals	Personalized health recommendations and follow-up support
Donation reminders	High	Donors forget future donation opportunities	Manual tracking by donors	Automated reminders and eligibility notifications
Emergency blood request communication	Medium	Delays in reaching suitable donors	Social media announcements	Targeted notifications and emergency alerts

Need	Priority	Concerns	Current Solution	Proposed Solution
Long-term donor engagement	Medium	Low donor retention after first donation	Limited incentive mechanisms	Gamification, achievements, donor tiers, and community features

3.5 Alternatives and Competition

Existing blood donation platforms in Vietnam are identified as LifeLine's primary competitors. These include App Hiến Máu, Giọt Máu Vàng Mobile Application, and Giọt Máu Vàng Website. Through survey analysis and platform evaluation, key characteristics are summarized as follows:

- **Giọt Máu Vàng Mobile Application:** Provides appointment scheduling, donation history tracking, QR identification. However, several functions are incomplete or act as placeholders, navigation is inconsistent, performance issues are frequently reported, and users must manually enter information despite document scanning support.
- **Giọt Máu Vàng Website:** Supports online appointment scheduling, donation history management, certificates, and donor information management. However, the website lacks responsive mobile design, does not provide push notifications, offers limited location visualization, lacks community interaction features, and requires manual data handling for several processes.
- **App Hiến Máu:** Provides blood donation registration, donation history tracking, notifications, and location searching. Nevertheless, the platform mainly functions as a standalone scheduling tool and lacks advanced engagement mechanisms, real-time hospital integration, and personalized donor support.

Survey results further indicate that users face difficulties accessing reliable donation information, are uncertain about donation eligibility, seek trustworthy post-donation health guidance, and show strong interest in reminders, emergency notifications, and AI-powered assistance. These findings reveal opportunities that are not fully addressed by existing platforms.

Through analyzing the limitations of current solutions and understanding user expectations, LifeLine aims to provide a comprehensive donor-centered ecosystem featuring AI-assisted guidance, personalized health support, community engagement, intelligent notifications, donor retention mechanisms, and stronger integration between donors, hospitals, and blood donation organizations.

4. Project Overview

(Author: Trịnh Khánh Linh | Reviewer: Trần Anh Kiệt | Editor: Trịnh Khánh Linh)

4.1 Product Perspective

LifeLine is a centralized blood donation management and donor engagement platform designed to connect donors, blood donation organizations, and hospitals within a single ecosystem.

The system enables users to discover donation opportunities, register for donation events, track their donation history, receive personalized health guidance, and stay informed through automated reminders and notifications. Healthcare organizations and hospitals can efficiently manage donor information, monitor donation activities, and communicate urgent blood requests when necessary.

By integrating management, communication, and engagement functionalities into a unified platform, LifeLine aims to improve operational efficiency while encouraging long-term donor participation.

4.2 Assumptions and Dependencies

The development team identifies several assumptions regarding system usage:

- Users have access to stable internet connections.
- Blood donation organizations and hospitals are willing to adopt digital management solutions.
- Donors possess basic digital literacy and can operate web or mobile applications.
- Email and notification services are available for communication purposes.
- AI-powered assistance services can be deployed and maintained reliably.

Some dependencies of the LifeLine platform are:

- Cloud-hosted database systems for storing donor and donation information.
- Email and notification services for reminders and alerts.
- Authentication and authorization services.
- AI services supporting donor assistance and eligibility guidance.
- Hospital and blood donation organization data sources for event and inventory management.

5. Product Features

5.1. User Features

5.1.1. User Account Management

(Author: Trần Anh Kiệt | Reviewer: Trịnh Khánh Linh | Editor: Trần Anh Kiệt)

No.	Feature	Description	Priority
1.1-1	**Registration via Citizen ID (CCCD)**	This feature allows new users to create a verified donor account by scanning their Citizen Identity Card (CCCD) QR code. The system automatically extracts and pre-fills personal data, including full name, date of birth, and ID number. Next, the user can enter additional personal information like email, phone number, password and receive a verification email to activate the account. This significantly reduces manual entry time and minimizes input errors during onboarding. Verification against a national identity document ensures that each donor profile is tied to a real, authenticated individual, eliminating duplicate accounts and maintaining the integrity of donation records. Blood centers and the platform as a whole benefit from a trustworthy, deduplicated donor registry, while donors enjoy a fast and frictionless registration experience. This feature is essential for establishing the foundational identity layer on which all other platform activities like bookings, history tracking, and emergency notifications depend.	High
1.1-2	**Login**	This feature provides donors with a secure and straightforward authentication mechanism using their CCCD number combined with a registered password. It ensures that only verified account holders can access personal donation records, appointment details, and health data stored in the system. A password reset flow via OTP verification (email) is also included to help users regain access without requiring staff intervention, ensuring that long-term users do not lose access to high-level accounts that hold significant sentimental value. In conclusion, this	High

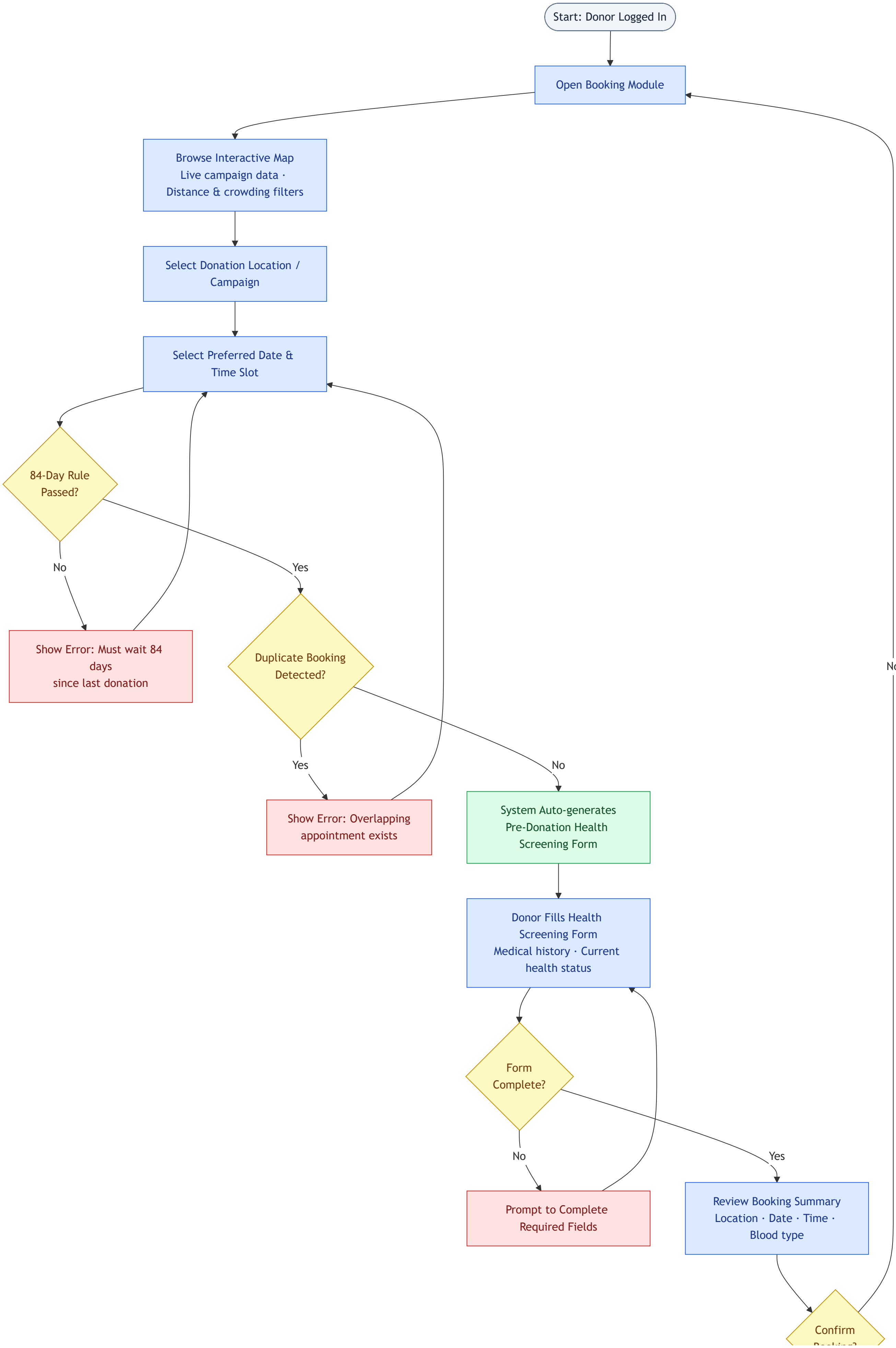
No.	Feature	Description	Priority
		benefits donors by protecting their sensitive medical information while keeping the login process familiar and accessible.	
1.1-3	**Profile Management**	This feature enables donors to view and update their personal profile at any time, including contact details such as phone number, email address, and residential address. Keeping profile information current is especially important for emergency alert targeting, where the system filters donors by location and blood type, and for appointment confirmations that rely on accurate contact channels. Donors benefit from a sense of ownership over their own data, while blood centers gain higher confidence in the accuracy of their communication outreach. The feature also displays key summary information such as blood type and donation schedule directly on the donor dashboard for quick reference. Overall, it ensures the platform can maintain an up-to-date and reliable donor database throughout the user's lifetime on the platform.	High

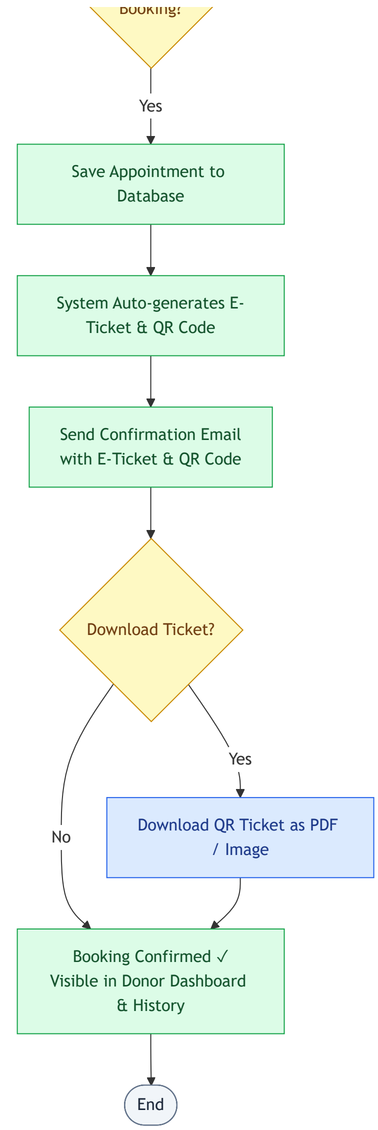
5.1.2. Donation Booking & Location Services

(Author: Trần Anh Kiệt | Reviewer: Trịnh Khánh Linh | Editor: Trần Anh Kiệt)

No.	Feature	Description	Priority
1.2-1	**Interactive Map and Location Discovery**	This feature integrates an interactive map interface that allows donors to visually explore nearby blood donation points and active campaigns in real time. Users can apply filters such as distance radius and crowding level to identify the most convenient and available donation location without manually browsing multiple information sources. The map pulls live data from the campaign database, ensuring that only currently active events and open venues are displayed. Donors benefit from a significant reduction in search effort and travel uncertainty, especially in urban areas where multiple centers operate simultaneously. Blood	High

No.	Feature	Description	Priority
		centers gain better attendance distribution across their events as donors are guided toward optimal locations.	
1.2-2	**Appointment Scheduling**	This feature enables donors to select their preferred date, time slot, and blood donation location, with the system automatically validating medical eligibility constraints before confirming the booking. Most critically, the system enforces the mandatory 84-day (12-week) waiting period between donations, blocking invalid bookings and displaying a clear error message if the constraint is violated ensuring full compliance with health and safety regulations. Upon successful scheduling, the system generates a personalized electronic appointment ticket encoded as a QR code and can also be downloaded in PDF or image format. Donors benefit from a streamlined, error-proof booking experience that removes the need for phone calls or in-person registration. Blood center staff benefit from reduced administrative overhead at check-in, as attendee data is pre-recorded and verifiable via QR scan.	High


Start: Donor Logged In
Open Booking Module
Browse Interactive Map
Live campaign data ·
Distance & crowding filters
Select Donation Location /
Campaign
Select Preferred Date &
Time Slot
84-Day Rule
Passed?
No
Yes
Show Error: Must wait 84
days
since last donation
Duplicate Booking
Detected?
Yes
No
Show Error: Overlapping
appointment exists
System Auto-generates
Pre-Donation Health
Screening Form
Donor Fills Health
Screening Form
Medical history · Current
health status
Form
Complete?
No
Yes
Prompt to Complete
Required Fields
Review Booking Summary
Location · Date · Time ·
Blood type
Confirm
No

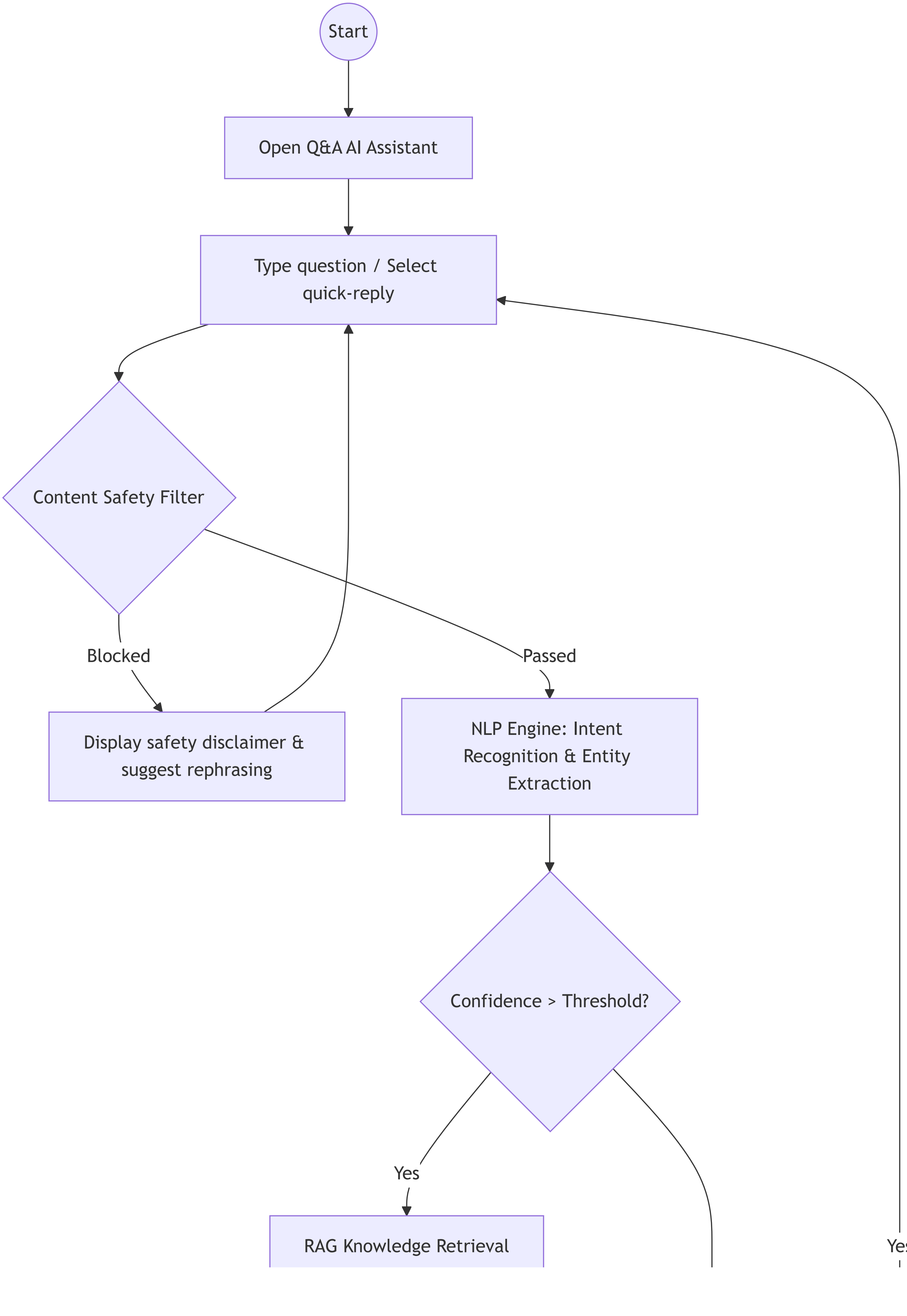


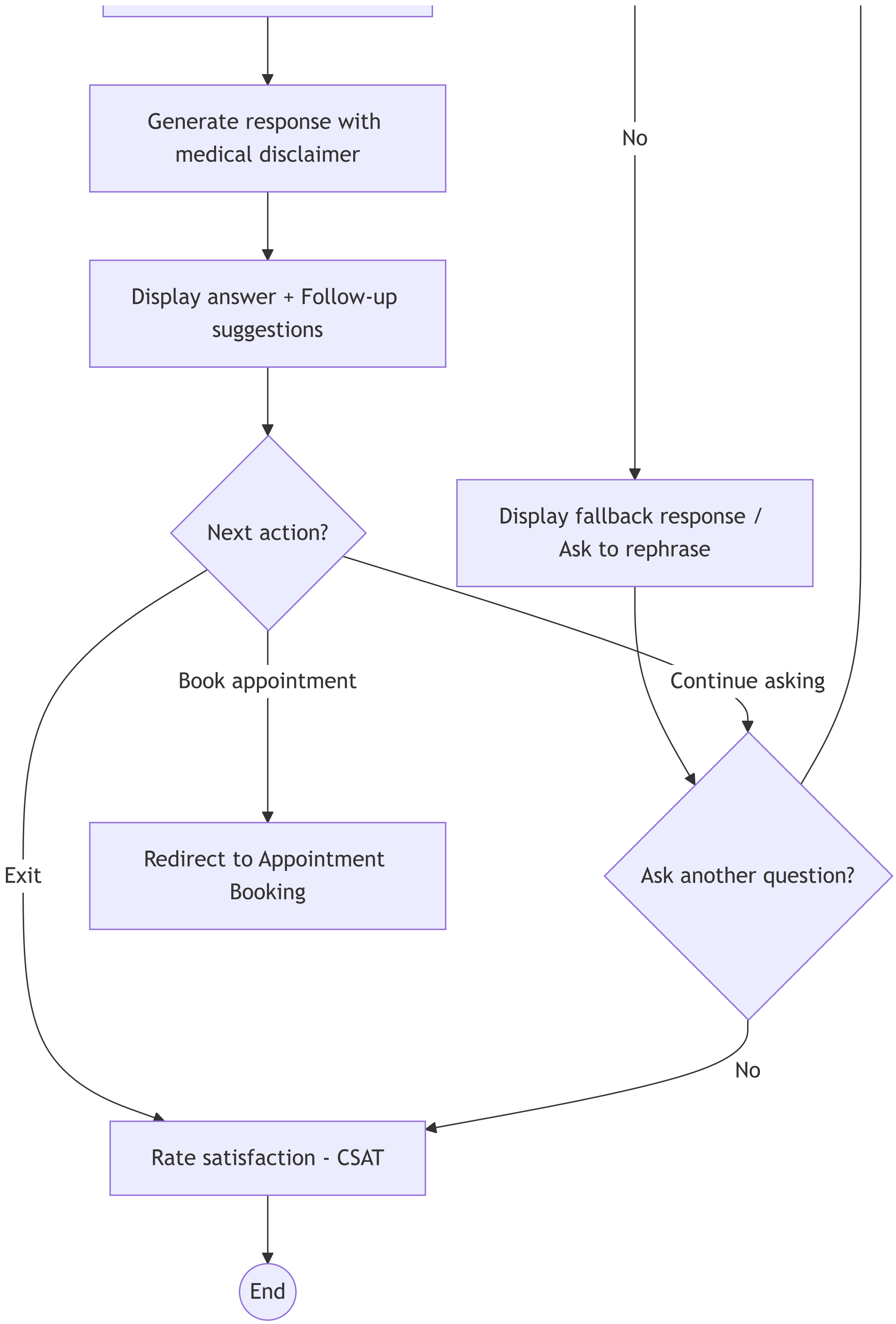
5.1.3 Donor Guidance and Support (Q&A AI)

(Author: Trần Đức Quý | Reviewer: Trần Anh Kiệt | Editor:Trần Đức Quý)

No.	Feature	Description	Priority
1.3-1	**AI-Powered Conversational Support & Guidance**	This feature introduces an intelligent AI assistant powered by a Retrieval-Augmented Generation (RAG) pipeline to serve as the primary self-service channel for donors. It handles multi-turn conversations to deliver instant, context-aware answers regarding donation procedures and provides tailored pre- and post-donation guidance. Additionally, the AI intelligently redirects eligible users to relevant platform features, such as appointment booking, to streamline the user journey. This ensures donors receive immediate, 24/7 support	High

No.	Feature	Description	Priority
		while significantly reducing the administrative burden on medical staff.	


Start
Open Q&A AI Assistant
Type question / Select quick-reply
Content Safety Filter
Blocked
Passed
Display safety disclaimer & suggest rephrasing
NLP Engine: Intent Recognition & Entity Extraction
Confidence > Threshold?
Yes
RAG Knowledge Retrieval
Yes



5.1.4 News, Notifications & Communication

(Author: Trần Đức Quý | Reviewer: Trần Anh Kiệt | Editor:Trần Đức Quý)

No.	Feature	Description	Priority
1.4-1	**Content Management System (CMS) & News Feed**	This feature acts as the platform's centralized engagement hub, equipping administrators with a robust Content Management System (CMS). Staff can seamlessly create and publish blood donation campaigns, health tips, and organizational updates directly to the platform's public feed. This keeps donors consistently informed and actively engaged with local donation activities. By centralizing communication, blood centers can ensure their outreach efforts are consistent, targeted, and widely accessible to the community.	Medium
1.4-2	**Automated Routine Notifications**	A multi-channel Notification Engine automatically dispatches time-sensitive routine alerts to donors based on their personalized user preferences. It delivers critical updates—such as appointment reminders and 84-day cycle unlocks—via email and web push notifications. This automated system minimizes missed appointments and prompts donors to take timely action without requiring manual follow-up from staff. Ultimately, it improves donor retention and engagement while actively preventing notification fatigue.	High
1.4-3	**SOS Emergency Broadcast System**	This feature provides a dedicated SOS Emergency Broadcast system that empowers verified hospitals to rapidly trigger urgent blood supply requests. The emergency system utilizes intelligent audience segmentation and dynamic geographic radius expansion to pinpoint the most suitable candidates. It instantly alerts these compatible, nearby donors during critical blood shortages. This highly targeted approach accelerates donor mobilization and streamlines life-saving emergency coordination between hospitals and blood centers.	High

5.1.5. Donation Impact & Tracking

(Author: Trịnh Khánh Linh | Reviewer: Trần Anh Kiệt | Editor: Trịnh Khánh Linh)

No.	Feature	Description	Priority
1.5-1	**Donation Timeline**	The Donation Timeline feature enables donors to view a chronological record of their entire blood donation journey within the platform. Users can access the timeline to review important milestones such as appointment registrations, eligibility confirmations, completed donations, recovery follow-ups, and achievement unlocks. The feature helps donors better understand their contribution history, improves transparency, and encourages continued participation in future donation activities.	Medium
1.5-2	**Milestone Badges & Achievements**	The Milestone Badges & Achievements feature enables the system to reward donors for reaching predefined contribution milestones. Users automatically receive digital badges after completing activities such as their first donation, multiple successful donations, emergency donations, or long-term participation goals. The feature helps recognize donor commitment, strengthen engagement, and encourage recurring donation behavior.	Medium
1.5-3	**Gamification Progress Tracking (Donor Levels)**	This feature enables donors to progress through a level-based reward system based on their participation and engagement within the platform. Users earn experience points through activities such as blood donations, campaign participation, profile completion, and community involvement to unlock higher donor levels. The feature helps create long-term motivation and makes the donation experience more engaging and rewarding.	Medium

5.1.6. Community

(Author: Trịnh Khánh Linh | Reviewer: Trần Anh Kiệt | Editor: Trịnh Khánh Linh)

No.	Feature	Description	Priority
1.6-1	**Direct Facebook Fanpage Link**	The Direct Facebook Fanpage Link feature enables users to quickly access the organization's official Facebook fanpage from within the platform. Users can visit the page to view announcements, campaign promotions, educational content, and community interactions. The feature helps organizations improve outreach while allowing donors to stay connected through a familiar communication channel.	Low

5.2. Blood Center Features

(Author: Trần Minh Triết | Reviewer: Trần Anh Kiệt | Editor: Trần Minh Triết)

5.2.1. Blood Donation Campaign and Management

No.	Feature	Description	Priority
2.1-1	**Event Creation and Configuration**	This feature enables blood center staff to create and manage blood donation campaigns through a centralized interface. Staff can configure important event details such as target blood groups, venue, schedule, and participant capacity to ensure that each campaign meets organizational requirements. The feature helps streamline campaign planning and prevents overbooking by enforcing registration limits. It benefits blood center administrators by reducing manual coordination efforts and improving event organization, while donors receive clearer and more reliable registration opportunities.	High
2.1-2	**QR Code Scanning and Verification**	The QR Code Scanning and Verification feature allows staff to quickly verify donor registrations by scanning the QR code contained in each donor's electronic ticket. Once scanned, the system instantly retrieves the donor's registration information and confirms eligibility for participation. This contactless process significantly reduces waiting time and minimizes human errors associated with manual check-in procedures. Both staff	High

No.	Feature	Description	Priority
		and donors benefit from a faster, more efficient, and more accurate verification experience during donation events.	
2.1-3	**Donor Registration Management**	This feature enables blood center staff to receive and manage donor registrations for each blood donation campaign. The system displays the current number of registrations, helps monitor event capacity, and automatically closes registration once the participant limit has been reached. Staff can view each donor's profile, including personal information, donation history, and screening results. In addition, the system allows staff to update donor statuses such as Eligible for Donation, Ineligible for Donation, or Donation Completed to reflect each donor's progress throughout the donation process. This feature helps blood centers control participant numbers, maintain centralized donor records, and improve coordination efficiency during blood donation campaigns.	High

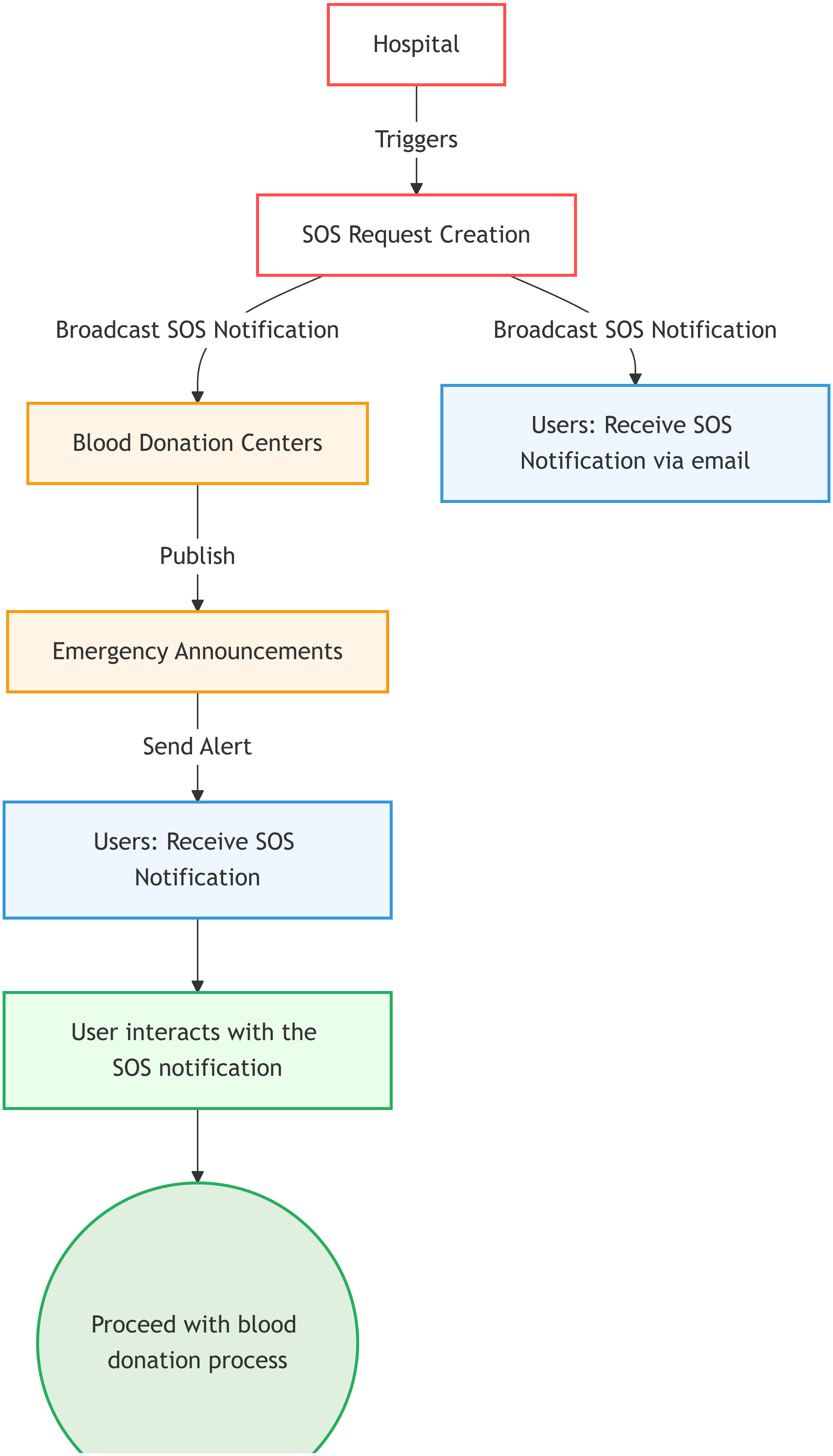
5.2.2. Communication and User Engagement Management

No.	Feature	Description	Priority
2.2-1	**Content Publishing**	This feature provides blood center staff with a built-in content management system for creating, editing, and publishing news articles, educational content, and campaign announcements. By centralizing communication within the application, donors can easily access the latest information regarding blood donation activities and organizational updates. The feature ensures that information remains consistent, up-to-date, and readily available to users. It benefits both blood centers, which can communicate more effectively, and donors, who stay informed about relevant opportunities and events.	Medium
2.2-2	**Emergency Announcements**	The Emergency Announcements feature enables blood centers to rapidly distribute urgent notifications when blood shortages or emergency situations occur. Staff	High

No.	Feature	Description	Priority
		can target specific donor groups based on criteria such as blood type, location, or donation eligibility to maximize the effectiveness of outreach efforts. This capability helps organizations respond quickly to critical demands and improve the availability of essential blood supplies. As a result, hospitals, blood centers, and patients all benefit from faster emergency coordination and donor mobilization.	

5.2.3. Blood Inventory and Emergency Coordination Management

No.	Feature	Description	Priority
2.3-1	**Blood Inventory and Emergency Coordination Management**	This feature enables blood center staff to comprehensively manage blood supplies stored in the inventory. The system allows staff to monitor detailed information for each blood bag, including blood type, quantity, collection date, expiration date, and storage location, while also providing statistical reports on available blood stock by blood group. In addition, staff can update inventory records whenever new blood units are added to the storage or existing units are dispatched for distribution. Through real-time inventory visibility and management, staff can effectively control blood supply levels and reduce shortages or waste caused by expired blood products. Blood centers, hospitals, and patients all benefit from more accurate, timely, and efficient blood inventory management and distribution.	High


Hospital
Triggers
SOS Request Creation
Broadcast SOS Notification
Broadcast SOS Notification
Blood Donation Centers
Users: Receive SOS Notification via email
Publish
Emergency Announcements
Send Alert
Users: Receive SOS Notification
User interacts with the SOS notification
Proceed with blood donation process

5.3 Hospital Features

(Author: Nguyễn Quốc Dương | Reviewer: Trần Anh Kiệt | Editor: Nguyễn Quốc Dương)

5.3.1 Emergency Blood SOS Request Management

No.	Feature	Description	Priority
3.1-1	**Emergency Blood Request Creation**	This feature enables hospital staff to create emergency blood requests when urgent blood supplies are needed for patients. Staff can specify the required blood type, quantity, urgency level, and expected timeframe for fulfillment. The feature provides a standardized process for communicating critical blood needs, reducing delays and misunderstandings during emergencies. Hospitals benefit from faster request submission, while blood centers and donors receive clear and consistent information regarding urgent blood demands.	High
3.1-2	**SOS Request Monitoring and Tracking**	This feature allows hospital staff to monitor the progress of emergency blood requests from creation to completion. Users can view request status, response activity, and overall fulfillment progress in a centralized location. Continuous visibility helps hospitals make informed operational decisions and respond proactively when shortages occur. The feature improves transparency and supports more effective emergency coordination among stakeholders.	High
3.1-3	**Emergency Request Reporting**	This feature provides historical records and summary reports related to emergency blood requests. Hospitals can review request outcomes, response effectiveness, and fulfillment statistics to evaluate operational performance. The information supports future planning and helps organizations identify opportunities for improvement. Both hospitals and blood centers benefit from greater insight into emergency blood coordination activities.	Medium

5.4 System Features (Backend & Automations)

(Author: Trần Minh Triết | Reviewer: Trần Anh Kiệt | Editor: Trần Minh Triết)

5.4.1. User-Facing Automations

No.	Feature	Description	Priority
4.1-1	**Pre-Donation Screening Form Generation**	This feature automatically generates a digital health screening form as part of the appointment booking flow, triggered immediately after a donor selects a campaign and time slot. The form collects current medical information including health status, medication history, recent travel and activity, and a consent declaration before the booking is finalized. By embedding the screening process directly into the registration workflow, the system ensures that eligibility assessment is completed before the donation day, eliminating the need for paper-based forms at the venue and significantly reducing administrative workload for blood center staff. Donors benefit from a guided, structured experience that surfaces ineligibility warnings in real time, while blood centers benefit from receiving already reviewed, standardized health data for every registered attendee. The system enforces business rules such as the 84-day waiting period and configurable ineligibility criteria, ensuring compliance with medical donation standards without requiring manual staff intervention.	High
4.1-2	**E-Ticket & QR Generation**	This feature automatically generates a personalized electronic appointment ticket encoded with a cryptographically signed, unique QR code immediately after a donation appointment is confirmed and saved to the database. The e-ticket compiles full appointment details such as donor name, blood type, donation date and time, location name and address, and a unique ticket ID which is showed to the donor on the website with the ticket attached as a PDF/Image, and a booking confirmation email will also be delivered to the donor's email to announce, the ticket is also accessible through the donor's appointment detail page. The QR code payload is digitally signed using asymmetric cryptography to prevent forgery,	High

No.	Feature	Description	Priority
		ensuring that only legitimate tickets can be verified at the donation venue by blood center staff using the QR scanning feature. Donors benefit from a contactless, paperless check-in experience that removes the need to bring printed documentation, while blood center staff benefit from faster, more accurate donor verification during high-attendance campaign events. This automation is tightly integrated with the downstream digital donor record generation and the staff QR scanning workflow, making it a foundational link in the end-to-end donation day pipeline.	

5.4.2. Blood Center-Facing Automations

No.	Feature	Description	Priority
4.2-1	**Digital Donor Record Generation**	This feature automatically creates a comprehensive digital donor registration record on the blood center side immediately after a donor's appointment is confirmed, consolidating data from the donor's profile, the completed pre-donation screening form, and the generated e-ticket into a single centralized document. The record includes a donor identity section, appointment details, all health screening responses with an auto-computed preliminary eligibility flag (Eligible / Requires Review), a real-time updatable donation status field (Registered -> Checked In -> Eligible -> Donation Completed / Ineligible), and a cross-linked QR ticket reference enabling instant record retrieval via the staff's QR scanning workflow. By generating this record automatically, the system eliminates manual data entry for blood center staff and ensures that a complete, organized donor roster is available before the campaign begins, covering every registered attendee. Blood center staff benefit from a structured, staff facing interface that supports the full donation day workflow from eligibility review at arrival to post-donation status updates and clinical note annotations without relying on paper forms or manual data compilation. The record is immediately visible in the	High

No.	Feature	Description	Priority
		campaign's donor roster view and remains stored in the database for lasting donor history tracking and audit purposes.	
4.2-2	**SOS Request Evaluation and Prioritization**	This feature automatically evaluates an incoming emergency blood request submitted and approved by hospital staff, performing real-time analysis across blood inventory records and the donor registry to identify the most suitable blood centers and eligible donors who can respond to the crisis. For blood center matching, the system ranks candidates by a composite score that considers available inventory volume of the required blood type, geographic proximity to the requesting hospital, and current dispatch capacity, ensuring that the most capable and accessible center is contacted first. For donor matching, the system queries the donor registry for compatible blood type, currently eligible donors within a configurable initial search radius, ranking them by proximity, time since last donation, and donor engagement tier; if insufficient donors are found, the radius is automatically expanded in increments until an adequate pool is identified. Emergency notifications are dispatched to the top ranked blood centers and donors within one minute of request approval via email and in-app push notification, bypassing standard rate limits to prioritize urgent outreach. Hospital staff benefit from rapid, coordinated mobilization of resources without manual coordination overhead, while blood centers and donors receive targeted, relevant alerts that minimize unnecessary notification volume and support the fastest possible emergency response.	High

5.4.3. Notification Service

No.	Feature	Description	Priority
4.3-1	**Emergency Alert**	This feature operates as the automated distribution engine for critical emergency blood requests, triggered immediately	High

No.	Feature	Description	Priority
	Broadcasting	after the SOS Request Evaluation and Prioritization process identifies the most suitable responders. Upon receiving the prioritized list of candidate blood centers and eligible donors, the system prepares urgent, highly visible alert payloads and dispatches them across multiple communication channels, including email, and in-app push notifications. To ensure immediate action during life-threatening shortages, these emergency broadcasts bypass standard notification queues and are delivered within one minute of request approval. By decoupling the evaluation logic from the notification delivery, the system guarantees a robust, highly available broadcasting mechanism. Ultimately, hospitals benefit from instantaneous, targeted mobilization without manual effort, while eligible donors and blood center staff receive distinct, high-priority alerts that cut through routine noise, empowering them to respond swiftly to critical community needs.	

5.5 Administrator Features

(Author: Trần Anh Kiệt | Reviewer: Trịnh Khánh Linh | Editor: Trần Anh Kiệt)

5.5.1. System and User Management

No.	Feature	Description	Priority
5-1	**User Account Management**	This feature provides administrators with full authority to view, create, update, suspend, or permanently delete user accounts across all roles within the platform. Administrators can search and filter accounts by role (Donor, Organization Staff, Hospital Staff), status (Active, Suspended), or registration date to efficiently locate specific users. The ability to manually activate or deactivate accounts ensures that access control remains reliable, particularly in cases where automated verification fails or accounts require corrective action. This feature is essential for maintaining a clean, trustworthy user database and preventing	High

No.	Feature	Description	Priority
		unauthorized or fraudulent access to the system. All account modification actions are logged with timestamps and administrator identity for full auditability.	
5-2	**Role and Permission Management**	This feature allows administrators to define, assign, and revoke roles and their associated permissions for all system users. Administrators can configure which actions each role (Donor, Organization Staff, Hospital Staff) is authorized to perform, ensuring that users can only access functionality relevant to their responsibilities. Role-based access control is critical for protecting sensitive data such as blood inventory records and personal donor information from unauthorized modification or disclosure. When organizational changes occur, such as a staff member changing position, administrators can update role assignments immediately without system downtime. This centralized permission layer underpins the security architecture of the entire platform.	High
5-3	**System Activity Monitoring**	This feature gives administrators real-time and historical visibility into system events, including user logins, failed authentication attempts, data modifications, and emergency alert broadcasts. Activity logs are searchable and filterable by user, action type, time range, and affected resource, enabling administrators to quickly identify suspicious behavior or operational anomalies. Proactive monitoring helps detect potential security threats such as unauthorized access attempts or unusual data access patterns before they escalate into serious incidents. The monitoring dashboard also provides usage statistics such as active sessions, peak usage periods, and feature adoption rates, supporting informed decisions about system scaling and maintenance. All logs are immutable and retained for audit and compliance purposes.	High
5-4	**System Configuration Management**	This feature enables administrators to manage platform-wide configuration settings, including notification parameters, eligibility rule thresholds (such as the 84-day	Medium

No.	Feature	Description	Priority
		donation interval), campaign registration limits, and integration endpoints for external services such as email providers and mapping APIs. Centralizing configuration through an admin interface rather than requiring code changes allows the platform to adapt quickly to evolving operational requirements or regulatory updates without system redeployment. Administrators can also manage content moderation settings, toggle feature availability, and configure backup schedules to ensure system reliability. This feature reduces dependency on technical development resources for routine operational adjustments and improves the platform's long-term maintainability.	

6. Non-Functional Requirements

(Author: Nguyễn Quốc Dương | Reviewer: Trần Anh Kiệt | Editor: Nguyễn Quốc Dương)

The following non-functional requirements apply to all features within the SOS Hospital Emergency Request functional group.

6.1 Performance Requirements

ID	Requirement	Priority
NFR-P-01	The system shall process an emergency blood request submission within 5 seconds under normal operating conditions.	High
NFR-P-02	The system shall identify potential blood centers and eligible donors within 30 seconds after a valid SOS request is submitted.	High
NFR-P-03	Emergency notifications shall be delivered to recipients within 1 minute after request approval.	High
NFR-P-04	The system shall support at least 10,000 registered users and 10 concurrent active users without significant performance degradation.	High
NFR-P-05	User-facing pages shall load within 3 seconds for 95% of requests under normal network conditions.	Medium

6.2 Security Requirements

ID	Requirement	Priority
NFR-S-01	All user authentication data shall be encrypted during transmission using HTTPS/TLS.	High
NFR-S-02	The system shall enforce role-based access control for Donors, Hospital Staff, Blood Center Staff, and Administrators.	High
NFR-S-03	Personal information and medical-related records shall only be accessible to authorized users.	High
NFR-S-04	All SOS request activities shall be logged for auditing and traceability purposes.	High
NFR-S-05	User sessions shall automatically expire after 30 minutes of inactivity.	Medium

6.3 Reliability and Fault Tolerance Requirements

ID	Requirement	Priority
NFR-R-01	The system shall maintain at least 99.5% service availability excluding scheduled maintenance periods.	High
NFR-R-02	Daily automated backups of critical system data shall be performed.	High
NFR-R-03	The system shall recover critical services within 30 minutes following a server failure.	High
NFR-R-04	Emergency request records shall not be lost due to application failures or unexpected interruptions.	High

6.4 Usability Requirements

ID	Requirement	Priority
NFR-U-01	The system shall provide a responsive user interface for desktop, tablet, and mobile devices.	High

ID	Requirement	Priority
NFR-U-02	The user interface shall support both English and Vietnamese languages.	Medium
NFR-U-03	Emergency alerts shall be visually distinguishable from standard notifications.	High
NFR-U-04	The system shall be accessible through modern web browsers including Chrome, Edge, Firefox, and Safari.	High

6.5 Applicable Standards

ID	Requirement	Priority
NFR-STD-01	The system shall fully comply with the Vietnamese Personal Data Protection Decree regarding the collection, storage, and processing of donor Citizen ID (CCCD) and medical screening data.	High
NFR-STD-02	All web communications shall conform to HTTPS/TLS security standards.	High
NFR-STD-03	Date, time, and timestamp formats shall follow ISO 8601 standards.	Medium
NFR-STD-04	System APIs shall follow RESTful API design principles and JSON data exchange standards.	Medium
NFR-STD-05	User interface components shall follow WCAG 2.1 Level AA accessibility guidelines where applicable.	Medium